MODBUS Protocol Manual

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DIRECTORY

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1 Data transfer mode

MODBUS for data transmission by two ways:RTU and ASCII.RTU mode use the 8 bits binary characters and ASCII mode use the 7 bits ASC characters.To be the ASCII mode transmission byte,it should separates the one byte of four highers and four lowers in RTU mode into two bytes.

Note: According to the characteristics of two transmission models, the anti-interference ability of ASCII is stronger.So ASCII mode is recommended;Baud rate is recommended to use more than 9600 BPS for RTU mode; The data digits must be 8 for ASCII mode without calibration.

RTU transmission mode of data frame adopts CRC check,and ASCII transmission mode adopts LRC check.

The difference of the two transmission modes:

Transfer mode	**ASCII (7 bit)**	**RTU (8 bit)**
Coding format	ASCII code (‘0’-‘9’ ‘A’-‘F’)	8bit binary (0x00 – 0xff)
Start bit Data bit Check bit Stop bit	1 7,8 Non,odd,even 1,2	1 8 Non,odd,even 1,2
The frame check	LRC	CRC16

2 Registers and Data types

Register type	length of the data	Register number	Description
COIL	1 bit	-	Boolean variable(ON/OFF)
FLOAT	32 bit	2	32 bit floating point Numbers(IEEE754 format)
INT	16 bit	1	Unsigned integer(0x0–0xFFFF)
DOUBLE	64 bit	4	64 bit floating point Numbers(IEEE754 format)

2.1 COIL

Boolean variable 0xFF00 -> ON 0x0000 -> OFF

2.2 FLOAT

Using two registers to store the floating point number of single precision IEEE754 format.

Each float contains 4 bytes:

SEEEEEEE EMMMMMMM MMMMMMMM MMMMMMMM

S:The sign bit 0->anode 1->cathode (1 byte)

E:exponent (8 bytes)

M:The decimal part of tail (23 bytes)

For example,0xC1480000 = -12.5

2.3 DOUBLE

Using four registers to store the floating point number of single precision IEEE754 format.

Each float contains 8 bytes:

S:The sign bit 0->anode 1->cathode (1 byte)

E:exponent (11 bytes)

M:The decimal part of tail (52 bytes)

2.4 INT

Using a storage register to store an unsigned integer.

For example,0x0025 = 37 0x1234 = 4660

3 Definition of data frame format

3.1 CMD=0x03(Read one or more registers)

This example data is the removed(address:0x0030) data frame of small signal to be read, instrument address = 1.

Request frame: PC - > instrument

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	03	30 33
Register starting address high	00	30 30
Register starting address low	30	33 30
Register the number of high	00	30 30
Register the number of low bytes	02	30 32
The frame check	C4 04	43 41
Packet Tail	NONE	0D 0A

Response frame: instrument - > PC

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	03	30 33
length of the data	04	30 34
Register 0x0030 data high bytes	00	30 30
Register 0x0030 data low bytes	00	30 30
Register 0x0031 data high bytes	3F	33 46
Register 0x0031 data low bytes	00	30 30
The frame check	EB C3	42 39
Packet Tail	NONE	0D 0A

The data of removed small signal returned to the response frame data is 0.5.

3.2 CMD=0x05(Write a COIL variable)

This example data is the removal of total data frames, instrument address = 1.

Request frame: PC - > instrument

Name of the data fields	**RTU** Sample	**ASC** Sample

Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	05	30 35
Register starting address high	00	30 30
Register starting address low	90	39 30
COIL variable high bytes	FF	46 46
COIL variable low bytes	00	30 30
The frame check	8C 17	36 42
Packet Tail	NONE	0D 0A

Response frame: instrument - > PC

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	05	30 35
Register starting address high	00	30 30
Register starting address low	90	39 30
COIL variable high bytes	FF	46 46
COIL variable low bytes	00	30 30
The frame check	8C 17	36 42
Packet Tail	NONE	0D 0A

3.3 CMD=0x06(Write a single register)

The example data is data frames of a flow unit=m3/h,instrument address = 1.

Request frame: PC - > instrument

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	06	30 36

Register starting address high	00	30 30
Register starting address low	00	30 30
Variable high bytes	00	30 30
Variable low bytes	00	30 30
The frame check	89 CA	46 39
Packet Tail	NONE	0D 0A

Response frame: instrument - > PC

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	06	30 36
Register starting address high	00	30 30
Register starting address low	00	30 30
Variable high bytes	00	30 30
Variable low bytes	00	30 30
The frame check	89 CA	46 39
Packet Tail	NONE	0D 0A

3.4 CMD=0x10(Write multiple register)

The example data is data frame of damping time = 0.1s ,Instrument address=1

Request frame: PC - > instrument

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	10	31 30
Register starting address high	00	30 30
Register starting address low	20	32 30
Register number high bytes	00	30 30
Register number low bytes	02	30 32

Length of the data	04	30 34
Written to register 0x0020 high	CC	43 43
Written to register 0x0020 low	CD	43 44
Written to register 0x0021 high	3D	34 44
Written to register 0x0021 low	CC	43 43
The frame check	4F DD	32 37
Packet Tail	NONE	0D 0A

Response frame: instrument - > PC

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31
Function codes	10	31 30
Register starting address high	00	30 30
Register starting address low	20	32 30
Register number high bytes	00	30 30
Register number low bytes	02	30 32
The frame check	40 02	43 44
Packet Tail	NONE	0D 0A

3.5 Fault return frame

Because of that can not be correctly complete the operation request frame, the instrument will return a report of fault code returns the frame, with the specific reason that can't complete the operation.

Such as the instantaneous flow unit will set to m3, but the instrument can't support the flow unit m3, so it will return the following fault response frame.

Fault response frame: instrument - > PC

Name of the data fields	**RTU** Sample	**ASC** Sample
Packet Header	NONE	3A
Instrument address	01	30 31

Function codes	86	38 36
Fault codes	43	34 33
The frame check	03 91	39 31
Packet Tail	NONE	0D 0A

Notes:1 Function codes of Fault response frame is the Function codes+0x80 of Request frame

2 Please refer to the specific fault code with Appendix 1: fault codes

4 Data frame calibration algorithm

4.1 LRC check

// LRC check range:from“Instrument address”to the byte before LRC frame check code

```
void LRC(unsigned char *buf, unsigned int len)
{
unsigned int i;
LRC = 0;
for (i=0; i<len; i++)
{
LRC += buf[i];
}
LRC = 0xff - LRC;
LRC++;
}
```

4.2 CRC16 check

```
const unsigned char    TAB_CRC_H[] = {
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
```

0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40,0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,
0x00,0xC1,0x81,0x40,0x01,0xC0,0x80,0x41,0x01,0xC0,0x80,0x41,0x00,0xC1,0x81,0x40
};

const unsigned char TAB_CRC_L[] = {

0x00,0xC0,0xC1,0x01,0xC3,0x03,0x02,0xC2,0xC6,0x06,0x07,0xC7,0x05,0xC5,0xC4,0x04,
0xCC,0x0C,0x0D,0xCD,0x0F,0xCF,0xCE,0x0E,0x0A,0xCA,0xCB,0x0B,0xC9,0x09,0x08,0xC8,
0xD8,0x18,0x19,0xD9,0x1B,0xDB,0xDA,0x1A,0x1E,0xDE,0xDF,0x1F,0xDD,0x1D,0x1C,0xDC,
0x14,0xD4,0xD5,0x15,0xD7,0x17,0x16,0xD6,0xD2,0x12,0x13,0xD3,0x11,0xD1,0xD0,0x10,
0xF0,0x30,0x31,0xF1,0x33,0xF3,0xF2,0x32,0x36,0xF6,0xF7,0x37,0xF5,0x35,0x34,0xF4,
0x3C,0xFC,0xFD,0x3D,0xFF,0x3F,0x3E,0xFE,0xFA,0x3A,0x3B,0xFB,0x39,0xF9,0xF8,0x38,
0x28,0xE8,0xE9,0x29,0xEB,0x2B,0x2A,0xEA,0xEE,0x2E,0x2F,0xEF,0x2D,0xED,0xEC,0x2C,
0xE4,0x24,0x25,0xE5,0x27,0xE7,0xE6,0x26,0x22,0xE2,0xE3,0x23,0xE1,0x21,0x20,0xE0,
0xA0,0x60,0x61,0xA1,0x63,0xA3,0xA2,0x62,0x66,0xA6,0xA7,0x67,0xA5,0x65,0x64,0xA4,
0x6C,0xAC,0xAD,0x6D,0xAF,0x6F,0x6E,0xAE,0xAA,0x6A,0x6B,0xAB,0x69,0xA9,0xA8,0x68,
0x78,0xB8,0xB9,0x79,0xBB,0x7B,0x7A,0xBA,0xBE,0x7E,0x7F,0xBF,0x7D,0xBD,0xBC,0x7C,
0xB4,0x74,0x75,0xB5,0x77,0xB7,0xB6,0x76,0x72,0xB2,0xB3,0x73,0xB1,0x71,0x70,0xB0,
0x50,0x90,0x91,0x51,0x93,0x53,0x52,0x92,0x96,0x56,0x57,0x97,0x55,0x95,0x94,0x54,
0x9C,0x5C,0x5D,0x9D,0x5F,0x9F,0x9E,0x5E,0x5A,0x9A,0x9B,0x5B,0x99,0x59,0x58,0x98,
0x88,0x48,0x49,0x89,0x4B,0x8B,0x8A,0x4A,0x4E,0x8E,0x8F,0x4F,0x8D,0x4D,0x4C,0x8C,
0x44,0x84,0x85,0x45,0x87,0x47,0x46,0x86,0x82,0x42,0x43,0x83,0x41,0x81,0x80,0x40

```
};

// CRC check range:from“Instrument address”to the byte before CRC frame check
code
void CRC(unsigned char *buf, unsigned int len)
{
unsigned int i;
unsigned char CRC_H , CRC_L , index, ch;
      CRC_H = 0xff;
CRC_L = 0xff;
for (i=0; i<len; i++)
{
ch = buf[i];
index = CRC_H ^ ch;
            CRC_H = CRC_L ^ TAB_CRC_H[index];
CRC_L = TAB_CRC_L[index];
}
}
```

5 Instrument variable address definition

Instrument address definition is divided into two parts: One, the even read address (for the address group, some version of the flow meter can't be accessed); Two, can't even read address(for compatibility with previous communication mode) .All of the read instructions are 03, writing instructions are 06 and 16, and the data format are unsigned short, long, float, double (that is when the flow meter send data to the PLC or configuration software, the low word will be the first one).

5.1 The even read address

Address(decimalism)	Address(hexadecimal)	Data format	Register definition
2000	0x7D0	float	The instantaneous flow floating-point

2002	0x7D2	float	The instantaneous velocity of floating-point
2004	0x7D4	float	Flow percentage floating-point
2006	0x7D6	long	Integral part of positive cumulative numerical
2008	0x7D8	float	Fractional part of positive cumulative numerical
2010	0x7DA	long	Integral part of reverse cumulative numerical
2012	0x7DC	float	Fractional part of reverse cumulative numerical
2014	0x7DE	unsigned short	The instantaneous flow unit
2015	0x7DF	unsigned short	The total unit
2016	0x7E0	unsigned short	Upper limit alarm status (1: alarm, 0: don't alarm)
2017	0x7E1	unsigned short	Lower limit alarm status (1: alarm, 0: no alarm)
2018	0x7E2	unsigned short	Blank pipe alarm status (1: alarm, 0: no alarm)

2019	0x7E3	float	Damping time
2021	0x7E5	float	Small signal excision (%)
2023	0x7E7	float	Upper limit of the frequency
2025	0x7E9	float	Pulse equivalent
2027	0x7EB	unsigned short	Impulse interference wave level
2028	0x7EC	float	Pulse width
2030	0x7EE	float	High alarm percentage (%)
2032	0x7F0	float	Low alarm percentage (%)
2034	0x7F2	float	Density (kg/m3)
2036	0x7F4	float	Range (m3)

For the writing operations, it will only support1,2 and 4 length.4 length is only allowed to write the total flow(integer part and decimal part will be written at the same time).

5.2 Information list

The following is support instrument data variable information list, the data are for HEX type.

The variable name	Register address	Length of the registers	Read the instructions	Written the instructions
COIL type				
Clear the total	0091	***	***	05
INT type				
The instantaneous flow unit(See appendix 2)	0000	0001	03	06

The instantaneous flow decimal digits(See appendix 3)	0008	0001	03	06
The total unit(See appendix 2)	0010	0001	03	06
Amount of decimal digits(See appendix 3)	0018	0001	03	06
Impulse level	0050	0001	03	06
Device address	0078	0001	03	06
Communication protocol(See appendix 3)	0058	0001	03	***
Baud rate(See appendix 3)	0060	0001	03	***
Data bits(See appendix 3)	0068	0001	03	***
Check(See appendix 3)	0070	0001	03	***
LONG type				
The total extension	0500	0002	03	***
Total	0502	0002	03	***
FLOAT type				
Damping time	0020	0002	03	10
Small signal excision (%)	0030	0002	03	10
Upper limit of frequency	0038	0002	03	10
Pulse equivalent	0040	0002	03	10
Pulse width	0048	0002	03	10
Low alarm (percentage)	00C0	0002	03	10
High alarm (percentage)	00D0	0002	03	10
Fluid density	0098	0002	03	10
The instantaneous flow rate	0708	0002	03	***
The total	0504	0002	03	***

DOUBLE type				
Range (20 mA)	0028	0004	03	10
Coefficient of sensor	0088	0004	03	10
The total	0090	0004	03	***
The instantaneous flow rate	0700	0004	03	***

6 Appendix 1: fault codes

0x01	Invalid instruction code
0x02	Register address is invalid
0x30	Parameters of the ultra limit
0x31	Parameters of super lower limit
0x32	Parameter option error
0x40	Invalid register length
0x41	Register does not support the current instruction code
0x42	The function of register is not specified
0x43	The instantaneous flow unit does not exist
0x44	The total unit does not exist
0x45	The highest frequency output exceeds the upper limit
0x46	The lowest frequency output exceeds the limit
0x47	The highest velocity exceeds the upper limit
0x48	Duty ratio exceeds the upper limit
0xFE	Data frame chaos
0xFF	Data frame validation errors

7 Appendix 2: common units definition

The instantaneous flow rate	m3/h	0
	m3/m	1
	m3/s	2

	L/h	3
	L/m	4
	L/s	5
	USG/h	6
	USG/m	7
	USG/s	8
	kg/h	9
	kg/m	10
	kg/s	11
	t/h	12
	t/m	13
	t/s	14
cumulants	L	0
	m3	1
	USG	2
	kg	3
	t	4

8 The appendix 3: symbol definition

Modbus Communication mode	RTU	0
	ASCII	1
Check way	Odd check	0
	Even check	1
	Non check	2
Baud rate	1200bps	0
	2400bps	1
	4800bps	2
	9600bps	3

	19200bps	4
	38400bps	5
Data bits	7 bits	0
	8 bits	1
Decimal digits	0 decimal places	0
	1 decimal places	1
	2 decimal places	2
	3 decimal places	3

9 Appendix 4: diameter symbol definition

Diameter	Code	Diameter	Code
DN1	0	DN600	27
DN1.5	1	DN700	28
DN2	2	DN750	29
DN3	3	DN800	30
DN4	4	DN900	31
DN5	5	DN1000	32
DN6	6	DN1100	33
DN8	7	DN1200	34
DN10	8	DN1300	35
DN15	9	DN1350	36
DN20	10	DN1400	37
DN25	11	DN1500	38
DN32	12	DN1600	39
DN40	13	DN1700	40
DN50	14	DN1800	41
DN65	15	DN2000	42
DN80	16	DN2100	43
DN100	17	DN2200	44
DN125	18	DN2300	45

DN150	19	DN2400	46
DN200	20	DN2500	47
DN250	21	DN2600	48
DN300	22	DN2700	49
DN350	23	DN2800	50
DN400	24	DN2900	51
DN450	25	DN3000	52
DN500	26		